

C Tasks

Week 4

Q1: 1-check if number is even or odd using bitwise operators (check for both negative and positive numbers).

2- swap two numbers using bitwise operators without using a temporary variable (don't use any condition statement or any type of loops).

3- Write a function to check if a number is a power of 2 using bitwise operations and counts the number of set bits (1s) in an integer (you may use call by reference).

4- Implement a function to find the only non-repeating element in an array where every other element appears twice. (you should use bitwise operators).

Q2: 1- Given an integer x, write a function that reverses its bits.

Ex input: $(11)_{10}$ which is $(1011)_2$ output $(13)_{10}$ which is $(1101)_2$

2- implement bitwise multiplication using only $\ll$, $\gg$, $\&$, and $+$.

(time is very important to be as small as possible)

3- implement an algorithm to find the missing number in an array of size n containing numbers from 0 to n using XOR?

hint (numbers in array are not repeated and

Q3: 1-Write a macro-SQUARE(x) that returns the square of a number. Use it in the main () function to print the square of 5 and square of 12.31

2-Define a macro SWAP (a, b, type) that swaps two variables of any type. hint (it will be function like macro with more than one line)

3-Write a macro Max_arr (arr, size) that takes an array of integer and its size and returns the maximum number in array.

Q4: 1-write header file that include typedefs of all types in C (signed and unsigned).

2- Solve question in week 3 Q2 but you must use typedefs of structure and use data types as you have written in the header file above.

Q5: 1-Explain the difference between malloc(), calloc(), realloc(), and free().

2-Write a function that dynamically allocates memory for a string, copies a given string into it, and returns the pointer.

3-Implement a program that dynamically allocates memory for a modified 2D array (matrix) of integers using malloc() , you will take number of rows and for every row you will take number of columns and the data of every row.

4-implement function that returns the transpose of the given matrix in question Q5.3 then print it and free this located memory. (make the columns to be the rows). (You must free all dynamic memory you have allocated)

Output samples of Q5.3 and Q5.4

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
enter number of rows: 3
enter number of columns in 1th row :2
enter elements of this row 1 2
enter number of columns in 2th row :3
enter elements of this row 3 4 5
enter number of columns in 3th row :4
enter elements of this row 6 7 8 9
matrix is
1 2
3 4 5
6 7 8 9
matrix transpose is
1 3 6
2 4 7
5 8
9
Process returned 0 (0x0)   execution time : 3
Press any key to continue.
```

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
enter number of rows: 3
enter number of columns in 1th row :2
enter elements of this row 1 2
enter number of columns in 2th row :1
enter elements of this row 3
enter number of columns in 3th row :2
enter elements of this row 4 5
matrix is
1 2
4 5
matrix transpose is
1 3 4
2 5
Process returned 0 (0x0)   execution time : 12.744 s
Press any key to continue.
```

Bonus (must solve one of these three questions)

Q1: Implement a function that finds the two non-repeating elements in an array where every other element appears twice. Hint (Using bitwise operators).

Q2: Define a macro IS_PRIME(n) the same as that in MATLAB that checks if a number is prime and returns 1 if true, 0 otherwise. (n is integer)

Hint (every number could be given by multiplication of number lower and number above its square root).

The solution must be with minimum possible time.

Explanation of prime numbers:

https://www.mathsisfun.com/prime_numbers.html

Q3: Solve question in week 3 Q2 but the number of student is not constant so there will be variable number of students.

Output sample of bonus Q3:

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
enter name of th1 student : ahmed
enter th1 student degrees in order (Arabic,English,math) 10 20 30
do you want to continue ? y
enter name of th2 student : hassan
enter th2 student degrees in order (Arabic,English,math) 70 80 100
do you want to continue ? y
enter name of th3 student : ali
enter th3 student degrees in order (Arabic,English,math) 17 13 30
do you want to continue ? y
enter name of th4 student : osama
enter th4 student degrees in order (Arabic,English,math) 90 50 13
do you want to continue ? y
enter name of th5 student : Khalad
enter th5 student degrees in order (Arabic,English,math) 90 80 80
do you want to continue ? n
the students sorted in increasing order based on sum of degrees :
ahmed
ali
osama
Khalad
hassan

Process returned 0 (0x0)   execution time : 82.573 s
Press any key to continue.
```